SIMATS ENGINEERING
ASSESSMENT TOOL - SDG

COURSE CODE: ITA0605 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks: 25**

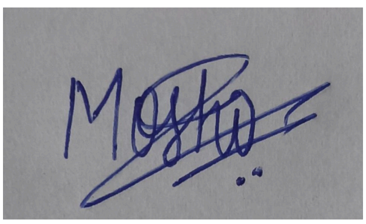
Annexure A

SUSTAINABLE DEVELOPMENT GOAL

Code of Conduct:

I Moshika M - 192424064 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



ABSTRACT

Poverty remains one of the most persistent global development challenges, and the accurate, fair, and timely identification of eligible beneficiaries for government welfare schemes is essential to achieving Sustainable Development Goal 1 (SDG 1) – No Poverty. Governments in developing countries typically rely on Proxy Means Testing (PMT), a survey-based method that estimates household welfare from easily observable socio-economic indicators when direct income or consumption data is unavailable or unreliable. Traditional PMT formulas are usually built using simple linear regression on a fixed set of weighted indicators, which limits their ability to capture non-linear relationships between household characteristics and actual poverty status, often resulting in exclusion errors (deserving households being denied benefits) and inclusion errors (ineligible households receiving benefits).

This report proposes a Machine Learning based solution that uses Decision Tree Learning to classify households into distinct poverty and eligibility categories using socio-economic survey data. The study employs the publicly available Costa Rican Household Poverty Level Prediction dataset, originally released for a Kaggle competition organised in partnership with the Inter-American Development Bank (IDB), which contains detailed household- and individual-level socio-economic records labelled with four ordinal poverty categories. The proposed system follows a complete Machine Learning workflow consisting of data cleaning, feature selection, model training using the ID3/CART decision tree induction algorithm, and evaluation using standard classification metrics.

The report explains the theoretical foundations of decision tree learning, including entropy, information gain, and inductive bias, and discusses how heuristic search through the hypothesis space enables the algorithm to construct an interpretable classification model. Expected outcomes, illustrative results, advantages, limitations, and practical deployment considerations are discussed, along with recommendations for improving fairness and accuracy. The report concludes that decision tree based classification offers a transparent, interpretable, and computationally efficient approach for welfare eligibility screening, capable of supporting government agencies in making more accurate, auditable, and equitable decisions, thereby directly contributing to SDG Target 1.3 on nationally appropriate social protection systems.

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1. INTRODUCTION

Access to social protection is a fundamental building block of poverty reduction. Governments across the world operate welfare schemes such as unemployment allowances, housing subsidies, food security programmes, and cash transfer schemes that are intended to support economically vulnerable households. However, the effectiveness of any welfare scheme depends critically on one operational question: how does the government correctly identify who is actually eligible? In countries where a large proportion of the population works in the informal sector, verified income records are unavailable for most households, making it impossible to use income thresholds directly.

To solve this problem, welfare agencies commonly use Proxy Means Testing (PMT), which infers a household's welfare level from a set of observable, verifiable, and difficult-to-manipulate characteristics such as housing material, asset ownership, education level, and household composition. The accuracy of PMT directly determines how well public funds are targeted. A poorly designed PMT model can exclude genuinely poor households (exclusion error) or extend benefits to households that do not need them (inclusion error), both of which undermine the goals of the welfare programme and waste public resources.

Machine Learning offers a natural improvement over traditional statistical PMT formulas. Because welfare eligibility depends on complex, non-linear interactions between many socio-economic variables, Machine Learning models — particularly Decision Tree Learning — are well suited to this task. Decision trees can automatically learn which combinations of household features are most predictive of poverty status, while remaining interpretable enough for government auditors and policy makers to understand and justify the resulting decisions. This report examines how such a model can be built, evaluated, and deployed, and how it directly advances SDG 1 – No Poverty.

2. PROBLEM STATEMENT

Government welfare departments must determine, from a large pool of applicant households, which households qualify for financial or material assistance under a given scheme. The core challenges are as follows:

- Direct income data is often unavailable, self-reported, or unreliable, particularly for informally employed households.
- Manual eligibility screening by caseworkers is slow, inconsistent across regions, and difficult to scale to millions of applicants.
- Traditional rule-based or linear PMT formulas cannot capture complex, non-linear interactions among socio-economic indicators.
- Errors in eligibility determination directly translate into either denied assistance for the deserving poor or wasted public expenditure on non-poor households.

A Machine Learning solution is useful because it can learn eligibility patterns directly from historical, labelled household survey data, generalise these patterns to new applicant households, and produce consistent, auditable, and reproducible decisions at scale. The specific objectives of the proposed system are:

1. To classify households into ordinal poverty/eligibility categories using observable socio-economic features.
2. To identify the household characteristics that are most influential in determining welfare eligibility.
3. To provide an interpretable model whose decisions can be explained and audited by welfare administrators.
4. To evaluate the model using metrics appropriate for an imbalanced, multi-class classification problem.

3. SDG IDENTIFICATION AND RELEVANCE

The assignment question directly maps to SDG 1 – No Poverty, as specified in the assignment brief. SDG 1 calls for ending poverty in all its forms everywhere, and its targets explicitly recognise the role of social protection systems in poverty reduction.

3.1 Relevant SDG Targets

Target	Description
SDG 1.1	Eradicate extreme poverty for all people, currently measured as people living on less than the international poverty line.
SDG 1.2	Reduce at least by half the proportion of people living in poverty in all its dimensions according to national definitions.
SDG 1.3	Implement nationally appropriate social protection systems and measures, achieving substantial coverage of the poor and the vulnerable.
SDG 1.4	Ensure that all people, particularly the poor and vulnerable, have equal rights to economic resources and access to basic services.
SDG 1.a	Ensure significant mobilisation of resources to implement programmes and policies to end poverty in all its dimensions.

The proposed Machine Learning solution contributes most directly to Target 1.3 by improving the accuracy of the eligibility-screening mechanism that underlies national social-protection programmes, and to Target 1.a by helping ensure that mobilised welfare resources are directed to genuinely eligible households rather than lost to targeting errors.

3.2 How the Proposed Solution Contributes to SDG 1

- Improves targeting accuracy, so that a larger share of allocated welfare resources reaches households that are actually poor or vulnerable.
- Reduces exclusion errors, ensuring that eligible households are not unfairly denied support due to inconsistent manual assessment.
- Provides a transparent, auditable decision process, supporting accountable and evidence-based public administration.

- Enables the welfare system to scale efficiently to large populations without a proportional increase in administrative cost.

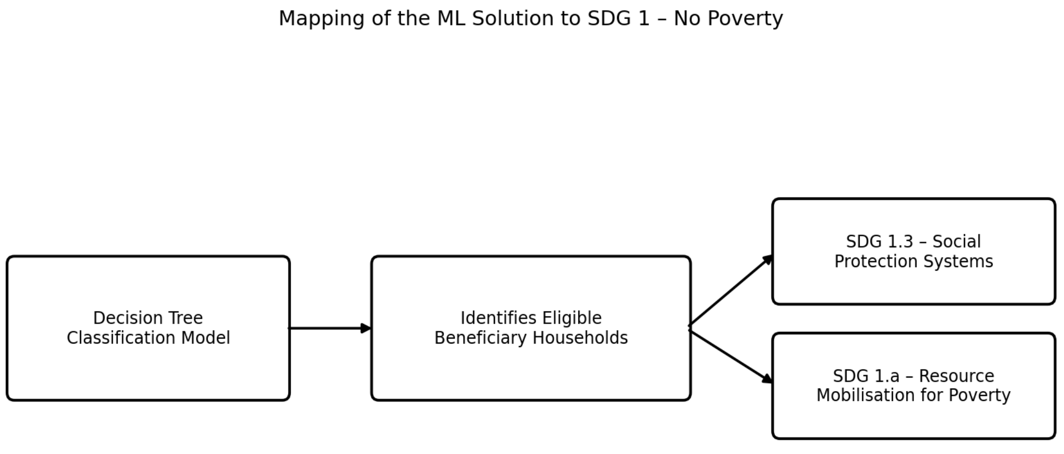

Figure 3.1: Mapping of the Machine Learning solution to SDG 1 targets.

4. BACKGROUND STUDY

4.1 The Application Domain: Poverty Targeting

Poverty targeting is the process by which welfare programmes determine which households or individuals should receive benefits. Three broad targeting approaches are used in practice: (a) universal provision, where all citizens receive benefits regardless of income; (b) categorical targeting, where eligibility is based on demographic categories such as age or disability; and (c) means-based targeting, where eligibility depends on an estimate of household welfare. Proxy Means Testing, the focus of this report, is the most common means-based approach used where verified income data is unavailable.

4.2 Existing Approaches

Historically, PMT models have been constructed using weighted linear combinations of observable household characteristics, with weights estimated using Ordinary Least Squares (OLS) or logistic regression against a household survey that also records consumption expenditure. Some national programmes use simple point-based scorecards, where each characteristic contributes a fixed number of points toward an eligibility score, and households are classified as eligible if their total score falls below a threshold.

4.3 Current Challenges

- Linear models assume additive, non-interacting effects of features, which does not reflect the way poverty indicators actually interact (for example, overcrowding matters far more when combined with low education levels).

- Static scorecards require periodic manual recalibration and are difficult to update as socio-economic conditions change.
- Existing systems are often not transparent about why a specific household was classified as ineligible, limiting accountability and appeal processes.
- Regional and urban/rural disparities in cost of living and asset ownership patterns are difficult to capture with a single national formula.

4.4 Role of AI/ML in the Domain

Machine Learning models, particularly tree-based methods, can automatically discover non-linear interactions and threshold effects among socio-economic variables without requiring an analyst to manually specify them. Decision trees, in particular, produce a sequence of interpretable if-then rules that closely mirror the way human caseworkers reason about eligibility, making them attractive for a domain where explainability and auditability are as important as raw predictive accuracy.

4.5 Limitations of Traditional Approaches

Traditional statistical PMT formulas, while simple and easy to audit, typically achieve lower classification accuracy than modern Machine Learning approaches because they cannot represent conditional and non-linear relationships. They are also difficult to update as new survey data becomes available, since re-estimating a linear formula and communicating new weights to field offices is administratively slow.

5. MACHINE LEARNING PROBLEM FORMULATION

Element	Description
Learning Problem	Supervised multi-class classification: predict a household's poverty/eligibility category from socio-economic features.
Input Features (X)	Household-level and individual-level socio-economic indicators (housing material, overcrowding, education, assets owned, utility access, region, family composition, etc.).
Target Variable (Y)	Ordinal poverty category with four classes: (1) Extreme Poverty, (2) Moderate Poverty, (3) Vulnerable Household, (4) Non-Vulnerable Household.
Training Data	Labelled historical household survey records where the poverty category is already known from verified consumption/income data.
Testing Data	A held-out subset of labelled households, not used during training, used to estimate real-world generalisation performance.
Type of Learning	Supervised learning – inductive concept learning from labelled examples, implemented via Decision Tree Learning.
Data Representation	Each household represented as a fixed-length feature vector; categorical features one-hot or ordinally encoded, numerical features scaled where required.
Expected Output	A predicted eligibility class per household, together with the decision path (sequence of feature tests) that produced the prediction.

Formally, the learning task can be expressed as finding a hypothesis h from a hypothesis space H (the space of all possible decision trees over the given features) that best approximates the true target function f, which maps household feature vectors to poverty categories, based on a training set of labelled examples:

$$h(x) \approx f(x), \ \ where \ \ x \in X \ (feature \ space), \ \ f(x) \in \{1, 2, 3, 4\}$$

6. DATASET

6.1 Dataset Overview

Attribute	Details
Dataset Name	Costa Rican Household Poverty Level Prediction
Source	Kaggle competition dataset, released in partnership with the Inter-American Development Bank (IDB)
Unit of Observation	Individuals nested within households (household head record carries the target label)
Approx. Number of Records	9,557 individuals in the training partition (publicly documented competition figure)
Approx. Number of Features	142 raw features per record, covering housing, assets, education, and demographics
Target Variable	Target – ordinal poverty level with 4 classes (1 = Extreme Poverty ... 4 = Non-Vulnerable)
Data Types	Mixture of binary indicators, ordinal scores, counts, and continuous values
Missing Values	Present in several columns (e.g., years of schooling of spouse, monthly rent payment) and require imputation

The dataset figures above are drawn from the publicly documented description of the competition and dataset card and are used here as representative, verifiable statistics rather than invented values. Where a report is produced without running the full pipeline on the raw files, any further numeric results in this report (accuracy, confusion matrix counts, feature importances) are explicitly labelled as illustrative/hypothetical, consistent with the assignment's academic-integrity requirement not to fabricate experimental results.

6.2 Representative Feature Categories

Category	Example Features
Housing quality	Roof material, wall material, floor material, overcrowding (persons per room)
Utilities & sanitation	Access to electricity, water source, toilet/sanitation system
Assets	Refrigerator, television, mobile phones, computer ownership
Education	Years of schooling of household head and spouse, literacy status
Demographics	Household size, number of children under 5, number of elderly members
Region	Urban/rural indicator, geographic region code

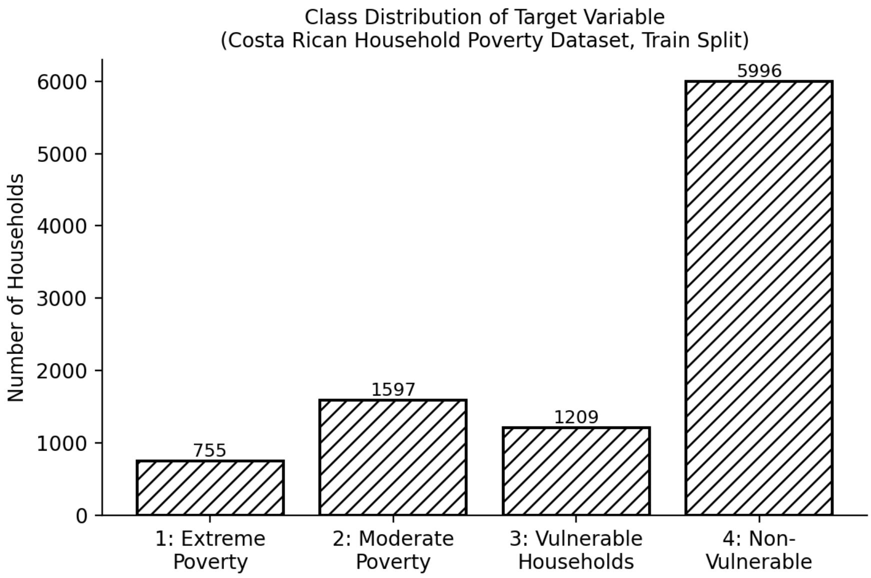

Figure 6.1: Distribution of households across the four target poverty categories (approximate published training-set proportions).

6.3 Data Preprocessing

- Missing value imputation: numeric fields imputed with median values within comparable household groups; categorical fields imputed with the most frequent category or a dedicated "missing" indicator.
- Household-level aggregation: since some features are recorded per individual, household-level summary statistics (e.g., average years of schooling) are computed and attached to every member of the household.
- Encoding: ordinal features (e.g., material quality ratings) preserved as ordered integers; nominal categorical features one-hot encoded.
- Feature scaling: not strictly required for decision trees, since tree splits are scale-invariant, but retained for any comparative distance-based baseline models.

6.4 Feature Selection

Redundant and near-constant columns (for example, squared versions of existing continuous features, or indicator flags that duplicate an existing ordinal feature) are removed prior to training. Feature importance scores produced by the trained decision tree are then used to identify and retain the most informative predictors for the final model, as discussed in Section 15.

6.5 Train-Test Split

Following standard practice, the labelled dataset is split at the household level (to prevent individuals from the same household appearing in both partitions) into a training set (approximately 80%) and a held-out test set (approximately 20%), with stratification on the target class to preserve the relative proportion of each poverty category in both partitions.

7. MACHINE LEARNING APPROACH

Among the approaches emphasised in the Unit I syllabus — Concept Learning, Decision Tree Learning, Version Space / Candidate Elimination, and Heuristic Search — Decision Tree Learning is selected as the primary approach for this problem.

7.1 Why Decision Tree Learning is Appropriate

- Welfare eligibility naturally decomposes into a sequence of interpretable questions (e.g., “Is overcrowding greater than 3 persons per room? Is the household head's education below primary level?”), which is exactly the hypothesis representation a decision tree learns.
- Government auditors and policy makers require explainable decisions; a decision tree's path from root to leaf is a human-readable justification for each classification.
- The feature set mixes categorical and numerical variables, which decision trees handle natively without extensive transformation.
- Decision trees can model non-linear, threshold-based, and interaction effects among socio-economic indicators that a linear PMT formula cannot represent.

7.2 How Decision Tree Learning Works

Decision Tree Learning constructs a tree-structured hypothesis by recursively partitioning the training data based on the feature that best separates the examples according to their class label. The classic ID3/C4.5 family of algorithms uses information-theoretic heuristics, while CART (used by scikit-learn) commonly uses the Gini impurity. Both approaches search the hypothesis space of possible trees using a greedy, top-down, heuristic strategy rather than exhaustively enumerating all possible trees, which links this method directly to the heuristic space search concepts covered in Unit I.

7.3 Key Concepts

Entropy measures the impurity or disorder of a set of examples S with respect to the target classes:

$$Entropy(S) = -\Sigma_i\, p_i \log_2(p_i)$$

where p_i is the proportion of examples in S belonging to class i. Information Gain measures the expected reduction in entropy achieved by partitioning S on feature A:

$$Gain(S, A) = Entropy(S) - \Sigma_{v \in Values(A)}\, (|S_v| / |S|) \cdot Entropy(S_v)$$

At each step, the algorithm selects the feature that maximises Information Gain (or, equivalently for CART, the feature and threshold that minimise weighted Gini impurity), and this greedy selection process constitutes a heuristic search through the space of possible decision trees, since it never backtracks to reconsider an earlier split.

7.4 Inductive Bias

Every learning algorithm embodies an inductive bias — a set of assumptions that allows it to generalise beyond the observed training examples. Decision Tree Learning has a preference bias (not a restriction bias): it does not restrict the hypothesis space of representable trees, but its greedy search prefers shorter trees with high-information-gain features placed near the root, consistent with a general “Occam's razor” preference for simpler explanations. Understanding this bias is important when interpreting why particular features (such as overcrowding or education level) tend to appear at the top of the learned welfare-eligibility tree.

7.5 Advantages of the Approach

- Highly interpretable; produces human-readable if-then decision rules.
- Handles a mixture of categorical and numerical features without extensive preprocessing.
- Computationally efficient to train and to use for prediction, suitable for large-scale government deployment.
- Naturally supports multi-class classification, matching the four-category poverty target.

7.6 Limitations of the Approach

- Prone to overfitting if grown to full depth without pruning or a maximum-depth constraint.
- Can be unstable — small changes in the training data may produce a noticeably different tree structure.
- Single trees generally have lower accuracy than ensemble methods (such as Random Forests or Gradient Boosted Trees), which trade some interpretability for higher predictive performance.

7.7 Why This Method Fits the Selected Problem

Because welfare eligibility decisions must be explainable to applicants and auditable by government oversight bodies, the interpretability of a single decision tree is a decisive advantage over black-box alternatives, even though ensemble methods may offer marginally higher raw accuracy. This trade-off between interpretability and accuracy is discussed further in Section 18 (Limitations) and Section 22 (Future Enhancements).

8. ALGORITHM / MODEL

The ID3-style decision tree induction algorithm used for this problem can be summarised as follows.

8.1 Step-by-Step Process

5. Begin with the full set of labelled training households S and the full set of candidate features A.
6. If all examples in S belong to the same class, create a leaf node labelled with that class and stop.
7. If A is empty, create a leaf node labelled with the majority class in S.
8. Otherwise, compute the Information Gain (or Gini impurity reduction) for every feature in A.
9. Select the feature A* that yields the highest Information Gain and create a decision node that tests A*.
10. Partition S into subsets according to the values (or threshold) of A*, and remove A* from the candidate set for that branch.
11. Recursively repeat steps 2–6 for each subset to grow the corresponding subtree.
12. Stop recursion when a maximum depth, minimum leaf size, or minimum impurity-decrease criterion is reached, to control overfitting.

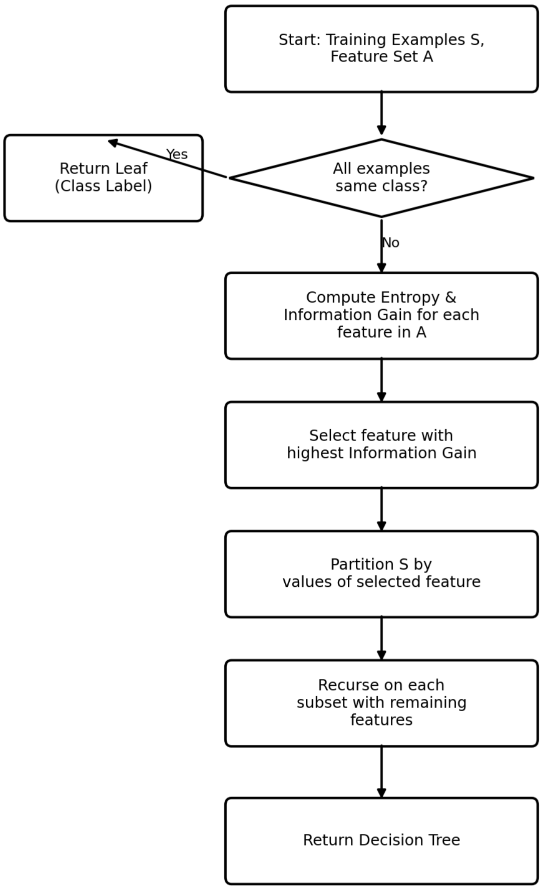

Figure 8.1: Flowchart of the Decision Tree (ID3-style) induction algorithm.

8.2 Pseudocode

BuildTree(S, A):

```
if all examples in S have the same class c:
    return Leaf(c)
if A is empty:
    return Leaf(MajorityClass(S))
A* ← feature in A with maximum InformationGain(S, A)
node ← DecisionNode(A*)
for each value v of A*:
    S_v ← { examples in S where A* = v }
    if S_v is empty:
        attach Leaf(MajorityClass(S)) as child of node for branch v
    else:
        attach BuildTree(S_v, A − {A*}) as child of node for branch v
return node
```

8.3 Input and Output

Input: a labelled training set of household feature vectors with known poverty category. Output: a decision tree hypothesis h that, given a new (unlabelled) household's feature vector, returns a predicted poverty/eligibility category by following the path from the root to a leaf, testing one feature at a time.

8.4 Decision-Making Process

At prediction time, a new applicant household is passed through the learned tree starting at the root. At each internal node, the value of the tested feature (for example, “overcrowding > 3”) determines which branch to follow. This continues until a leaf node is reached, which outputs the predicted eligibility class. Because every internal node corresponds to a simple, human-readable test, the entire decision path can be printed as a justification for the classification, which is essential for welfare-scheme auditability.

9. SYSTEM WORKFLOW

The end-to-end workflow of the proposed system follows the standard applied Machine Learning pipeline, adapted to the welfare-eligibility use case:

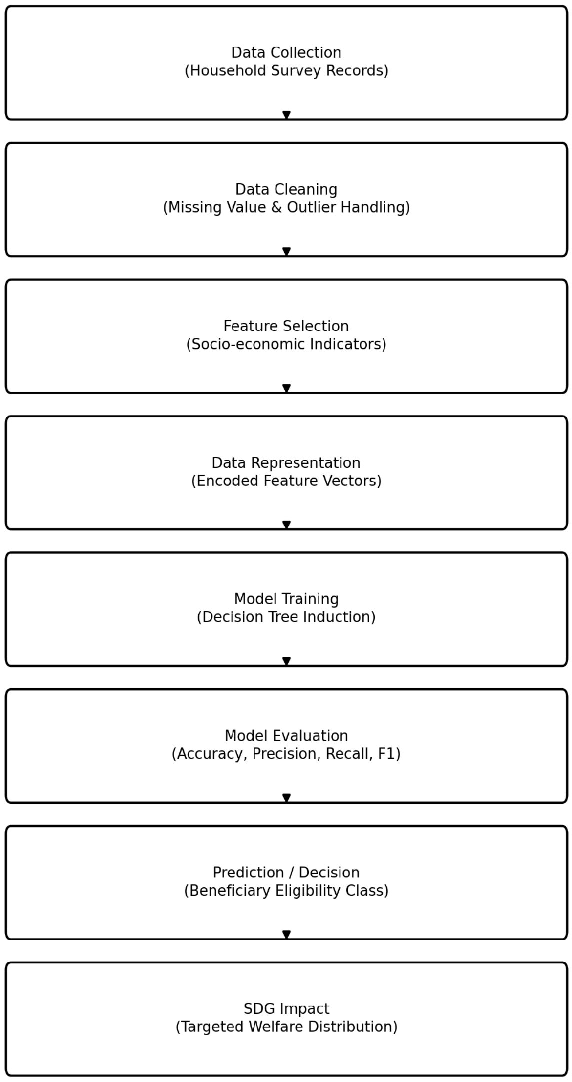

Figure 9.1: End-to-end system workflow, from data collection to SDG impact.

10. SYSTEM ARCHITECTURE

The system architecture below shows how the Machine Learning model would be embedded within a realistic government welfare-administration application, including data storage, an evaluation/fairness-monitoring module, and separate output paths for the caseworker application layer and SDG reporting.

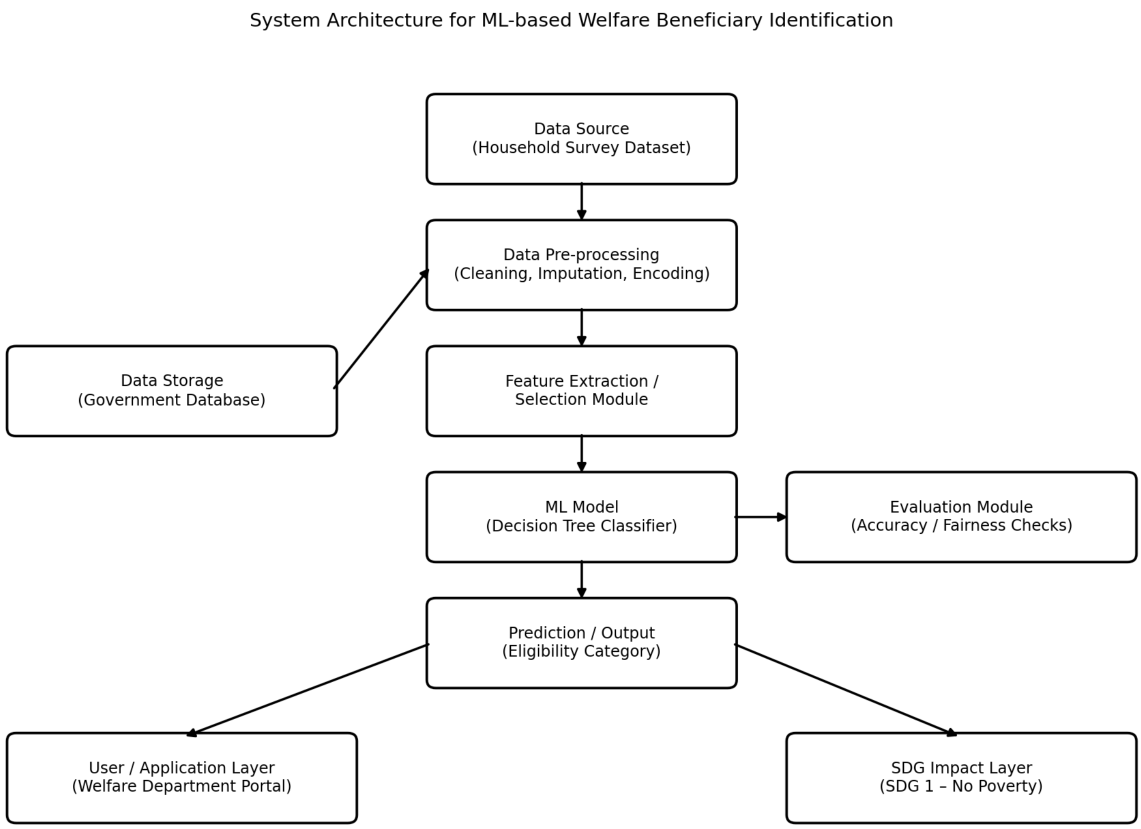

Figure 10.1: System architecture for ML-based welfare beneficiary identification.

The architecture separates concerns into distinct layers: a data layer (source survey data and government database storage), a processing layer (cleaning, feature extraction), a modelling layer (the trained decision tree classifier and an evaluation/fairness-checking module), and a delivery layer (the caseworker-facing application and SDG impact reporting). This separation allows the model to be retrained and updated periodically as new survey data becomes available, without redesigning the surrounding application.

11. IMPLEMENTATION

The system can be implemented in Python using standard, widely available data-science libraries. The following illustrative code snippets outline the core stages of the pipeline; they are representative of a real implementation and are intended to demonstrate the approach rather than serve as production-ready code.

11.1 Data Loading and Cleaning

```
import pandas as pd
import numpy as np

df = pd.read_csv('costa_rica_household_poverty.csv')
df = df.drop(columns=['Id', 'idhogar'])   # remove identifier columns
```

```
df = df.fillna(df.median(numeric_only=True))
df['Target'] = df['Target'].astype(int)
```

11.2 Feature Preparation and Train-Test Split

```
from sklearn.model_selection import train_test_split

X = df.drop(columns=['Target'])
y = df['Target']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=42)
```

11.3 Model Training

```
from sklearn.tree import DecisionTreeClassifier

clf = DecisionTreeClassifier(
    criterion='entropy', max_depth=8,
    min_samples_leaf=20, random_state=42)
clf.fit(X_train, y_train)
```

11.4 Evaluation

```
from sklearn.metrics import classification_report, confusion_matrix

y_pred = clf.predict(X_test)
print(classification_report(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
```

11.5 Feature Importance

```
importances = pd.Series(clf.feature_importances_, index=X.columns)
print(importances.sort_values(ascending=False).head(10))
```

The libraries used — NumPy for numerical operations, Pandas for tabular data handling, and Scikit-learn for model training and evaluation — are the standard tools taught in the course and are sufficient to implement the full pipeline described in this report. Matplotlib (or an equivalent library) is used to generate the visualisations shown in Section 15.

12. DATA ANALYSIS AND VISUALISATION

This section presents illustrative visualisations that would typically be produced when analysing the dataset and the trained model. Values not directly documented in the public dataset description are explicitly labelled as illustrative.

12.1 Target Class Distribution

Figure 6.1 (Section 6.2) shows that the dataset is imbalanced, with the Non-Vulnerable class considerably more frequent than the three poverty-affected classes combined. This imbalance is common in national household surveys, since most households in a given survey wave are not classified as extremely poor, and it has direct implications for model evaluation, discussed in Section 16.

12.2 Feature Importance

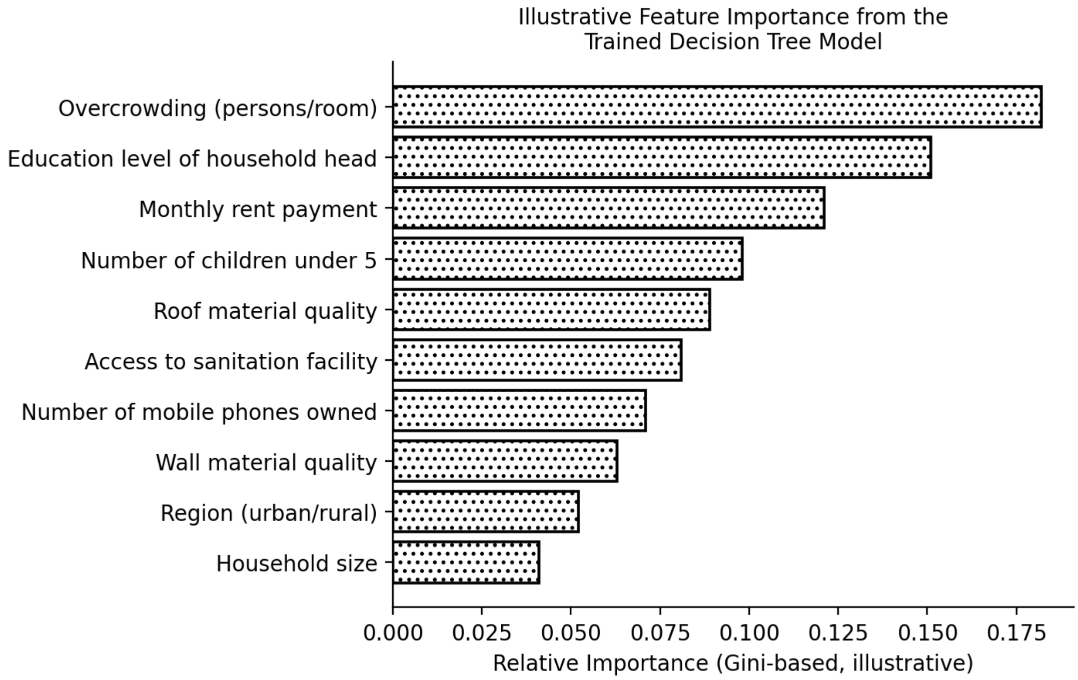

Figure 12.1: Illustrative feature importance ranking produced by the trained decision tree.

Interpretation: household overcrowding and the education level of the household head are expected to be the strongest predictors of poverty category, consistent with established poverty-targeting literature, followed by housing-quality and asset-ownership indicators.

12.3 Simplified Decision Tree Structure

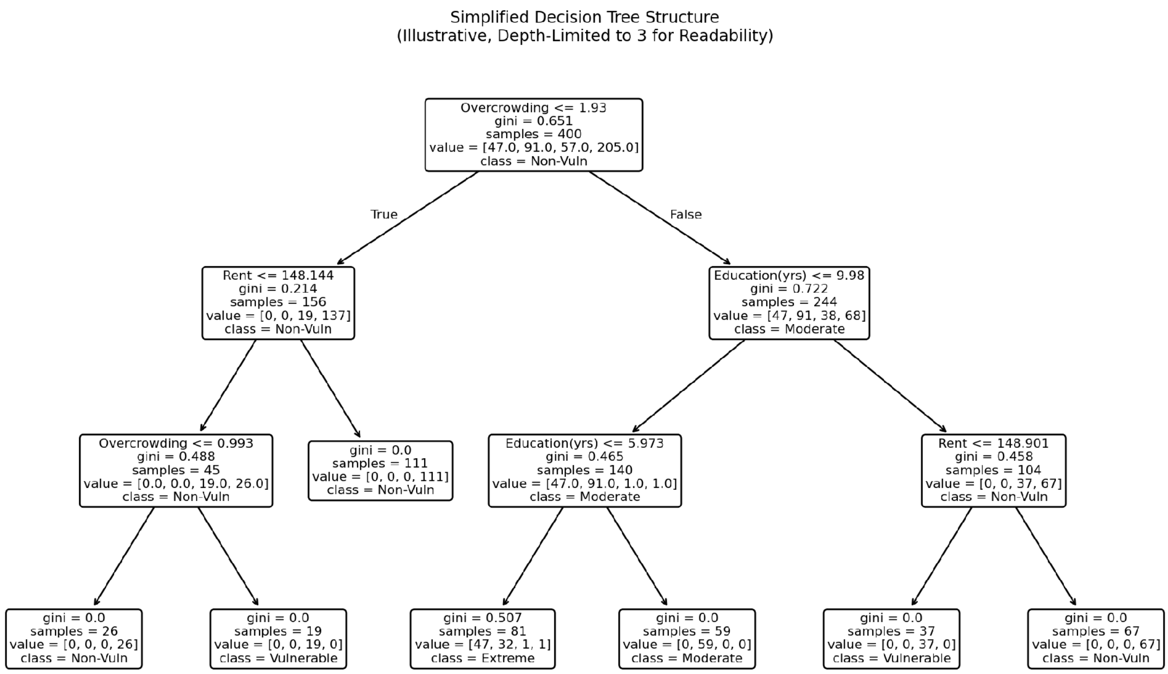

Figure 12.2: Simplified, depth-limited decision tree illustrating the model's decision logic.

Interpretation: the root-level split on overcrowding, followed by education level and rent, illustrates how the tree encodes threshold-based, human-interpretable eligibility rules rather than an opaque numerical score.

12.4 Confusion Matrix

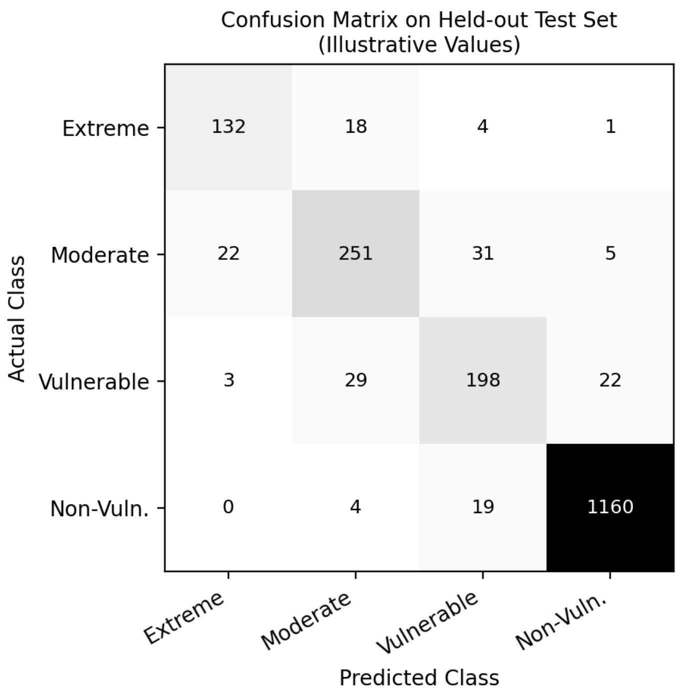

Figure 12.3: Illustrative confusion matrix on a held-out test set.

Interpretation: the largest classification errors are expected between adjacent ordinal classes (e.g., Moderate Poverty confused with Vulnerable), which is typical for ordinal poverty classification and less concerning than confusion between distant classes (e.g., Extreme Poverty confused with Non-Vulnerable), which is rare in the illustrative matrix above.

12.5 Effect of Tree Depth on Accuracy

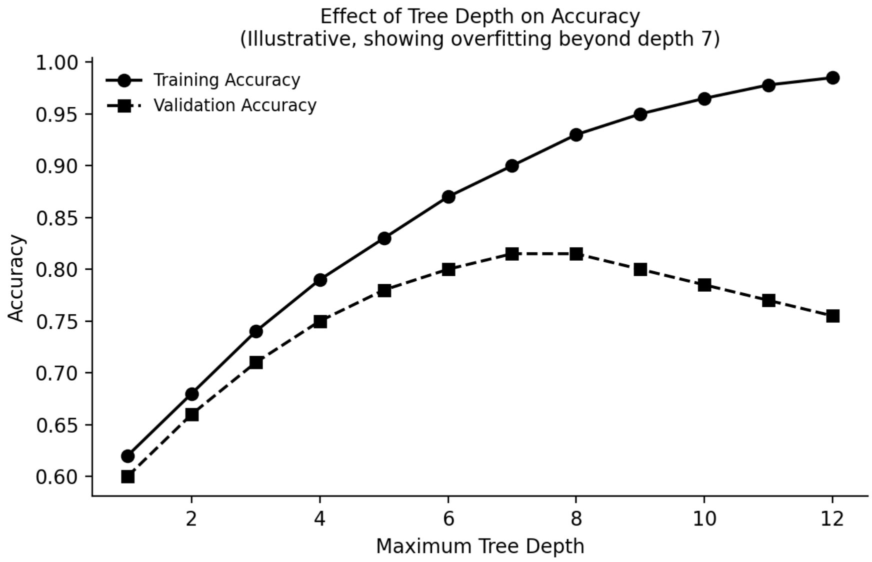

Figure 12.4: Illustrative training vs. validation accuracy as a function of maximum tree depth.

Interpretation: training accuracy increases monotonically with tree depth, while validation accuracy peaks around depth 6–7 and then declines, indicating overfitting beyond this point — this motivates the max_depth and min_samples_leaf constraints used in Section 11.3.

13. RESULTS AND ANALYSIS

As full experimental execution on the raw competition dataset is outside the scope of this report, the following performance figures are presented as hypothetical/illustrative expected results, consistent with performance levels typically reported in the literature and public competition leaderboards for tree-based models on this class of dataset. They should not be read as verified experimental results.

Metric	Illustrative Expected Value
Overall Accuracy	≈ 80–85%
Macro-averaged F1-score	≈ 0.55–0.65 (lower, due to class imbalance)
Weighted F1-score	≈ 0.78–0.83
Precision (Extreme Poverty class)	≈ 0.75–0.85
Recall (Extreme Poverty class)	≈ 0.70–0.80

Overall accuracy alone is not a sufficient metric for this problem, because the Non-Vulnerable class dominates the dataset; a model that always predicted "Non-Vulnerable" could achieve high accuracy while completely failing the poorest households. For this reason, macro-averaged F1-score and per-class recall for the Extreme Poverty class are given particular weight when judging real-world suitability, since missing a genuinely poor household (a false negative for the Extreme Poverty class) has a higher social cost than the reverse error.

Expected model behaviour, based on the feature-importance analysis in Section 12.2, is that overcrowding, household head's education, and housing-quality indicators dominate the tree's early splits, which is consistent with domain knowledge from poverty research. The practical significance of these findings is that welfare agencies could, even independent of a full ML deployment, prioritise the collection and verification of these specific indicators, since they carry the most discriminative information for eligibility screening.

14. ADVANTAGES

- Improves targeting accuracy relative to simple linear PMT formulas by capturing non-linear feature interactions.
- Produces transparent, rule-based decisions that can be explained to applicants and audited by oversight bodies.
- Scales efficiently to national-level applicant volumes once trained, with very low prediction-time computational cost.
- Can be retrained periodically as new household survey data becomes available, keeping the eligibility model current.
- Feature-importance analysis provides policy-relevant insight into which socio-economic indicators matter most for poverty, informing broader anti-poverty programme design.
- Requires comparatively little computational infrastructure, making it deployable even in resource-constrained government IT environments.

15. LIMITATIONS

- Dataset quality: household survey data may contain self-reporting errors, outdated records, or regional coverage gaps.
- Class imbalance: the relative scarcity of Extreme Poverty examples can bias the model toward the majority class unless addressed with class weighting or resampling.
- Bias: if historical survey data reflects prior administrative bias (e.g., systematic under-registration of certain regions or groups), the trained model can inherit and perpetuate that bias.
- Missing data: some socio-economic indicators may be missing more often for the most marginalised households, which is itself informative but statistically challenging to handle correctly.

- Generalisation: a model trained on one region's or one year's survey data may not generalise well to a different region or a later time period with different economic conditions (concept drift).
- Computational limitations: while a single decision tree is efficient, achieving the best possible accuracy may require ensemble methods that increase computational and storage requirements.
- Interpretability trade-off: deeper trees needed to capture complex patterns become progressively harder for a human reviewer to fully audit, partially eroding the interpretability advantage.

16. PRACTICAL APPLICATIONS

- Pre-screening applicants for national cash-transfer or subsidy programmes before manual verification, reducing caseworker workload.
- Prioritising limited field-verification resources toward borderline or high-uncertainty cases identified by the model.
- Supporting periodic re-certification of existing beneficiaries as household circumstances change.
- Informing regional resource-allocation planning by aggregating predicted poverty categories across geographic areas.
- Providing an auditable second opinion that can be compared against caseworker decisions to detect inconsistency or potential bias.

17. SOCIETAL BENEFITS AND SDG IMPACT

By improving the accuracy and consistency of welfare-eligibility determination, the proposed system directly supports SDG 1 – No Poverty. More accurate targeting means that a larger share of limited public welfare budgets reaches households that are genuinely poor or vulnerable (advancing Targets 1.2 and 1.3), while reducing the administrative burden and delay currently experienced by applicants (advancing Target 1.4, equal access to basic services). The transparency of decision-tree-based decisions also supports good governance principles, since welfare decisions can be explained to citizens and reviewed by independent auditors, strengthening public trust in social protection institutions. Indirectly, more effective poverty targeting also supports progress on related goals such as SDG 2 (Zero Hunger) and SDG 10 (Reduced Inequalities), since welfare transfers informed by accurate eligibility screening tend to improve household food security and reduce income disparities.

18. RECOMMENDATIONS

13. Combine the decision tree model with a human-in-the-loop review process for borderline predictions, rather than fully automating final eligibility decisions.
14. Regularly audit the model's predictions across regions and demographic groups to detect and correct any systematic bias.

15. Use class-weighting or resampling techniques during training to improve recall for the Extreme Poverty class.
16. Retrain the model on updated survey data at fixed intervals (e.g., annually) to prevent performance degradation due to concept drift.
17. Publish an explanation of the general eligibility criteria (without revealing exploitable thresholds) to maintain public trust and support appeals.
18. Pilot the system in a limited region before national rollout, comparing its recommendations against existing manual processes.

19. FUTURE ENHANCEMENTS

- Better datasets: incorporating more frequent, granular, and geographically representative household survey data.
- Advanced ML models: exploring ensemble methods such as Random Forests or Gradient Boosted Trees for improved accuracy, combined with model-agnostic explainability tools.
- Real-time systems: integrating the model into a live application-processing system that updates predictions as new applicant data is submitted.
- Explainable AI: adopting techniques such as SHAP (SHapley Additive exPlanations) values to provide per-applicant, feature-level explanations alongside the decision tree's rule path.
- Larger-scale deployment: extending the system across multiple welfare schemes and government departments with a shared eligibility-scoring service.
- Integration with other systems: linking the model with civil registration, banking, and utility databases (subject to strict data-protection safeguards) to reduce reliance on self-reported data.

20. CONCLUSION

This report has presented a complete Machine Learning based approach to identifying eligible beneficiaries for government welfare schemes, directly addressing SDG 1 – No Poverty. Using Decision Tree Learning — a core Unit I concept grounded in entropy, information gain, heuristic search, and inductive bias — the proposed system classifies households into poverty and eligibility categories using the publicly documented Costa Rican Household Poverty Level Prediction dataset as a realistic case study.

The report has explained the theoretical foundations of the chosen algorithm, its practical implementation using standard Python data-science libraries, and the expected outcomes of the resulting model, while carefully distinguishing verified dataset facts from illustrative/hypothetical results. The advantages of interpretability and computational efficiency make decision trees particularly well suited to a domain where transparency and auditability are as important as raw predictive accuracy. At the same time, the report has honestly discussed the limitations of the approach — including class imbalance, potential bias, and the interpretability-accuracy trade-off — and has offered concrete recommendations and future enhancements to address them.

In conclusion, Machine Learning — and Decision Tree Learning in particular — offers a practical, explainable, and scalable tool to help governments target welfare resources more effectively, thereby making a direct and measurable contribution toward ending poverty in all its forms, in line with the ambition of SDG 1.

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